

作業二

[Help](#)

Warning: The hard deadline has passed. You can attempt it, but **you will not get credit for it**. You are welcome to try it as a learning exercise.

☐ In accordance with the Coursera Honor Code, I (Sandipan Dey) certify that the answers here are my own work.

Question 1

Questions 1-2 are about noisy targets.

Consider the bin model for a hypothesis h that makes an error with probability μ in approximating a deterministic target function f (both h and f outputs $\{-1, +1\}$). If we use the same h to approximate a noisy version of f given by

$$P(\mathbf{x}, y) = P(\mathbf{x})P(y|\mathbf{x})$$

$$P(y|\mathbf{x}) = \begin{cases} \lambda & y = f(\mathbf{x}) \\ 1 - \lambda & \text{otherwise} \end{cases}$$

What is the probability of error that h makes in approximating the noisy target y ?

- ☐ μ
- ☐ $\lambda(1 - \mu) + (1 - \lambda)\mu$
- ☐ $1 - \lambda$
- ☒ $\lambda\mu + (1 - \lambda)(1 - \mu)$

Question 2

Following Question 1, with what value of λ will the performance of h be independent of μ ?

- ☐ 1
- ☒ 0.5
- ☐ 0

- ☐ 0 or 1

Question 3

Questions 3-5 are about generalization error, and getting the feel of the bounds numerically. Please use the simple upper bound $N^{d_{vc}}$ on the growth function $m_{\mathcal{H}}(N)$, assuming that $N \geq 2$ and $d_{vc} \geq 2$ }

For an $\mathcal{H}$ with $d_{vc} = 10$, if you want 95% confidence that your generalization error is at most 0.05, what is the closest numerical approximation of the sample size that the VC generalization bound predicts?

- ☐ 420,000
- ☐ 460,000
- ☐ 480,000
- ☐ 440,000

Question 4

There are a number of bounds on the generalization error ϵ , all holding with probability at least $1 - \delta$. Fix $d_{vc} = 50$ and $\delta = 0.05$ and plot these bounds as a function of N . Which bound is the tightest (smallest) for very large N , say $N = 10,000$?

- ☐ Original VC bound: $\epsilon \leq \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$
- ☐ Devroye: $\epsilon \leq \sqrt{\frac{1}{2N} (4\epsilon(1 + \epsilon) + \ln \frac{4m_{\mathcal{H}}(N^2)}{\delta})}$
- ☐ Parrondo and Van den Broek: $\epsilon \leq \sqrt{\frac{1}{N} (2\epsilon + \ln \frac{6m_{\mathcal{H}}(2N)}{\delta})}$
- ☐ Rademacher Penalty Bound: $\epsilon \leq \sqrt{\frac{2 \ln(2Nm_{\mathcal{H}}(N))}{N}} + \sqrt{\frac{2}{N} \ln \frac{1}{\delta}} + \frac{1}{N}$

Question 5

Continuing from Question 4, for small N , say $N = 5$, which bound is the tightest (smallest)?

- ☐ Devroye
- ☐ Rademacher Penalty Bound
- ☐ Parrondo and Van den Broek
- ☐ Original VC bound

Question 6

In Questions 6-11, you are asked to play with the growth function or VC-dimension of some hypothesis sets.

What is the growth function $m_{\mathcal{H}}(N)$ of "positive-and-negative intervals on $\mathbb{R}$ "? The hypothesis set $\mathcal{H}$ of "positive-and-negative intervals" contains the functions which are +1 within an interval $[ell, r]$ and -1 elsewhere, as well as the functions which are -1 within an interval $[ell, r]$ and +1 elsewhere.

- ☐ None of the other choices.
- ☐ $N^2 - N + 2$
- ☐ N^2
- ☐ $N^2 + 1$

Question 7

Continuing from the previous problem, what is the VC-dimension of the "positive-and-negative intervals on $\mathbb{R}$ "

- ☐ 3
- ☐ ∞
- ☐ 4
- ☐ 5

Question 8

What is the growth function $m_{\mathcal{H}}(N)$ of "positive donuts in $\mathbb{R}^2$ "? The

hypothesis set $\mathcal{H}$ of "positive donuts" contains hypotheses formed by two concentric circles centered at the origin. In particular, each hypothesis is +1 within a "donut" region of $a^2 \leq x_1^2 + x_2^2 \leq b^2$ and -1 elsewhere. Without loss of generality, we assume $0 < a < b < \infty$.

- ☐ None of the other choices.
- ☐ $N+1$
- ☐ $\binom{N+1}{3}+1$
- ☐ $\binom{N+1}{2}+1$

Question 9

Consider the "polynomial discriminant" hypothesis set of degree D on $\mathbb{R}$, which is given by
$$\mathcal{H} = \left\{ h_{\mathbf{c}} \mid h_{\mathbf{c}}(x) = \text{sign}\left(\sum_{i=0}^D c_i x^i\right) \right\}$$
 What is the VC-Dimension of such an $\mathcal{H}$?

- ☐ None of the other choices.
- ☐ $D+1$
- ☐ D
- ☐ ∞

Question 10

Consider the "simplified decision trees" hypothesis set on $\mathbb{R}^d$, which is given by
$$\mathcal{H} = \left\{ h_{\mathbf{t}, \mathbf{S}} \mid h_{\mathbf{t}, \mathbf{S}}(\mathbf{x}) = 2 \left[\mathbf{v} \in \mathbf{S} \right] - 1, \text{ where } \mathbf{v}_i = \left[x_i > t_i \right], \text{ and } \mathbf{S} \text{ a collection of vectors in } \{0, 1\}^d, \mathbf{t} \in \mathbb{R}^d \right\}$$
 That is, each hypothesis makes a prediction by first using the d thresholds t_i to locate $\mathbf{x}$ to be within one of the 2^d hyper-rectangular regions, and looking up $\mathbf{S}$ to decide whether the region should be +1 or -1. What is the VC-dimension of the "simplified decision trees" hypothesis set?

- ☐ $2^{d+1}-3$
- ☐ 2^d
- ☐ None of the other choices.
- ☐ ∞

Question 11

Consider the "triangle waves" hypothesis set on $\mathbb{R}$, which is given by

$$\mathcal{H} = \{ h_{\alpha} \mid h_{\alpha}(x) = \text{sign}(|\alpha x \bmod 4 - 2| - 1), \alpha \in \mathbb{R} \}$$
 Here $(x \bmod 4)$ is a number $x - 4k$ for some integer k such that $x - 4k \in [0, 4)$. For instance, $(11.26 \bmod 4)$ is 3.26, and $(-11.26 \bmod 4)$ is 0.74. What is the VC-Dimension of such an $\mathcal{H}$?

- ☐ None of the other choices.
- ☐ 2
- ☐ ∞
- ☐ 1

Question 12

In Questions 12-15, you are asked to verify some property or bounds on the growth function and VC-dimension.

Which of the following are upper bounds of the growth function $m_{\mathcal{H}}(N)$ for $N \geq d_{\text{vc}}$?

- ☐ $2^{d_{\text{vc}}}$
- ☐ $m_{\mathcal{H}}(\lfloor \frac{N}{2} \rfloor) \cdot m_{\mathcal{H}}(\lceil \frac{N}{2} \rceil)$
- ☐ $N^{d_{\text{vc}}} + 1$
- ☐ $\min_{1 \leq i \leq N-1} 2^i m_{\mathcal{H}}(N-i)$

Question 13

Which of the following are possible growth functions $m_{\mathcal{H}}(N)$ for some hypothesis set?

- ☐ $1 + N + \frac{N(N-1)(N-2)}{6}$
- ☐ 2^N
- ☐ $2^{\lfloor \sqrt{N} \rfloor}$
- ☐ $2^{\lfloor N/2 \rfloor}$

Question 14

For hypothesis sets $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_K$ with finite, positive VC dimensions $d_{\text{VC}}(\mathcal{H}_k)$, some of the following bounds are correct and some are not. Which among the correct ones is the tightest bound on the VC dimension of the intersection of the sets: $d_{\text{VC}}(\bigcap_{k=1}^K \mathcal{H}_k)$? (The VC dimension of an empty set or a singleton set is taken as zero)

- ☐ $0 \leq d_{\text{VC}}(\bigcap_{k=1}^K \mathcal{H}_k) \leq \max_{k=1}^K d_{\text{VC}}(\mathcal{H}_k)$
- ☐ $\min_{k=1}^K d_{\text{VC}}(\mathcal{H}_k) \leq d_{\text{VC}}(\bigcap_{k=1}^K \mathcal{H}_k) \leq \max_{k=1}^K d_{\text{VC}}(\mathcal{H}_k)$
- ☐ $0 \leq d_{\text{VC}}(\bigcap_{k=1}^K \mathcal{H}_k) \leq \sum_{k=1}^K d_{\text{VC}}(\mathcal{H}_k)$
- ☐ $0 \leq d_{\text{VC}}(\bigcap_{k=1}^K \mathcal{H}_k) \leq \min_{k=1}^K d_{\text{VC}}(\mathcal{H}_k)$

Question 15

For hypothesis sets $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_K$ with finite, positive VC dimensions $d_{\text{VC}}(\mathcal{H}_k)$, some of the following bounds are correct and some are not. Which among the correct ones is the tightest bound on the VC dimension of the union of the sets: $d_{\text{VC}}(\bigcup_{k=1}^K \mathcal{H}_k)$?

- ☐ $\max_{\{d_{\{vc\}}(\mathcal{H}_k) \}_{k=1}^K; \forall e; d_{\{vc\}}(\bigcup_{k=1}^K \mathcal{H}_k); \forall e; K-1 + \sum_{k=1}^K d_{\{vc\}}(\mathcal{H}_k)$
- ☐ $\max_{\{d_{\{vc\}}(\mathcal{H}_k) \}_{k=1}^K; \forall e; d_{\{vc\}}(\bigcup_{k=1}^K \mathcal{H}_k); \forall e; \sum_{k=1}^K d_{\{vc\}}(\mathcal{H}_k)$
- ☐ $0; \forall e; d_{\{vc\}}(\bigcup_{k=1}^K \mathcal{H}_k); \forall e; K-1 + \sum_{k=1}^K d_{\{vc\}}(\mathcal{H}_k)$
- ☐ $\min_{\{d_{\{vc\}}(\mathcal{H}_k) \}_{k=1}^K; \forall e; d_{\{vc\}}(\bigcup_{k=1}^K \mathcal{H}_k); \forall e; \sum_{k=1}^K d_{\{vc\}}(\mathcal{H}_k)$

Question 16

For Questions 16-20, you will play with the decision stump algorithm.

In class, we taught about the learning model of "positive and negative rays" (which is simply one-dimensional perceptron) for one-dimensional data. The model contains hypotheses of the form: $h_{\{s, \theta\}}(x) = s \cdot \text{sign}(x - \theta)$. The model is frequently named the "decision stump" model and is one of the simplest learning models. As shown in class, for one-dimensional data, the VC dimension of the decision stump model is 2.

In fact, the decision stump model is one of the few models that we could easily minimize E_{in} efficiently by enumerating all possible thresholds. Ties on the choice of threshold can be broken by choosing the median of equivalent thresholds; ties on the choice of s can be broken by choosing the arbitrary one. In this problem, you are asked to implement such an algorithm and run your program on an artificial data set. First of, start by generating a one-dimensional data by the procedure below:

(a) Generate x by a uniform distribution in $[-1, 1]$.

(b) Generate y by $f(x) = \tilde{s}(x) + \text{noise}$ where $\tilde{s}(x) = \text{sign}(x)$ and the noise flips the result with 20% probability.

For any decision stump $h_{\{s, \theta\}}$ with $\theta \in [-1, 1]$, express $E_{\text{out}}(h_{\{s, \theta\}})$ as a function of θ and s .

- ☐ $0.5 + 0.3s(|\theta| - 1)$
- ☐ $0.3 + 0.5s(|\theta| - 1)$
- ☐ $0.3 + 0.5s(1 - |\theta|)$
- ☐ $0.5 + 0.3s(1 - |\theta|)$

Question 17

Generate a data set of size 20 by the procedure above and run the one-dimensional decision stump algorithm on the data set. Record E_{in} and compute E_{out} with the formula above. Repeat the experiment 5,000 times. What is the average E_{in} ? Choose the closest option.

- ☐ 0.25
- ☐ 0.05
- ☐ 0.35
- ☐ 0.15

Question 18

Continuing from the previous question, what is the average E_{out} ? Choose the closest option.

- ☐ 0.25
- ☐ 0.15
- ☐ 0.35
- ☐ 0.05

Question 19

Decision stumps can also work for multi-dimensional data. In particular, each decision stump now deals with a specific dimension i , as shown below. $h_{\{s, i, \theta\}}(\mathbf{x}) = s \cdot \text{sign}(x_i - \theta)$. Implement the following decision stump algorithm for multi-dimensional data:

- a) for each dimension $i = 1, 2, \dots, d$, find the best decision stump $h_{\{s, i, \theta\}}$ using the one-dimensional decision stump algorithm that you have just implemented.
- b) return the "best of best" decision stump in terms of E_{in} , with ties arbitrarily broken.

The training data $\mathcal{D}_{\text{train}}$ is available at:

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw2%2Fhw2_train.dat

The testing data $\mathcal{D}_{\text{test}}$ is available at:

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw2%2Fhw2_test.dat

Run the algorithm on the $\mathcal{D}_{\text{train}}$. Report the E_{in} of the optimal decision stump returned by your program.

- ☐ 0.35
- ☐ 0.05
- ☐ 0.15
- ☐ 0.25

Question 20

Use the returned decision stump to predict the label of each example within the $\mathcal{D}_{\text{test}}$. Report an estimate of E_{out} by E_{test} .

- ☐ 0.35
- ☐ 0.15
- ☐ 0.05
- ☐ 0.25

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